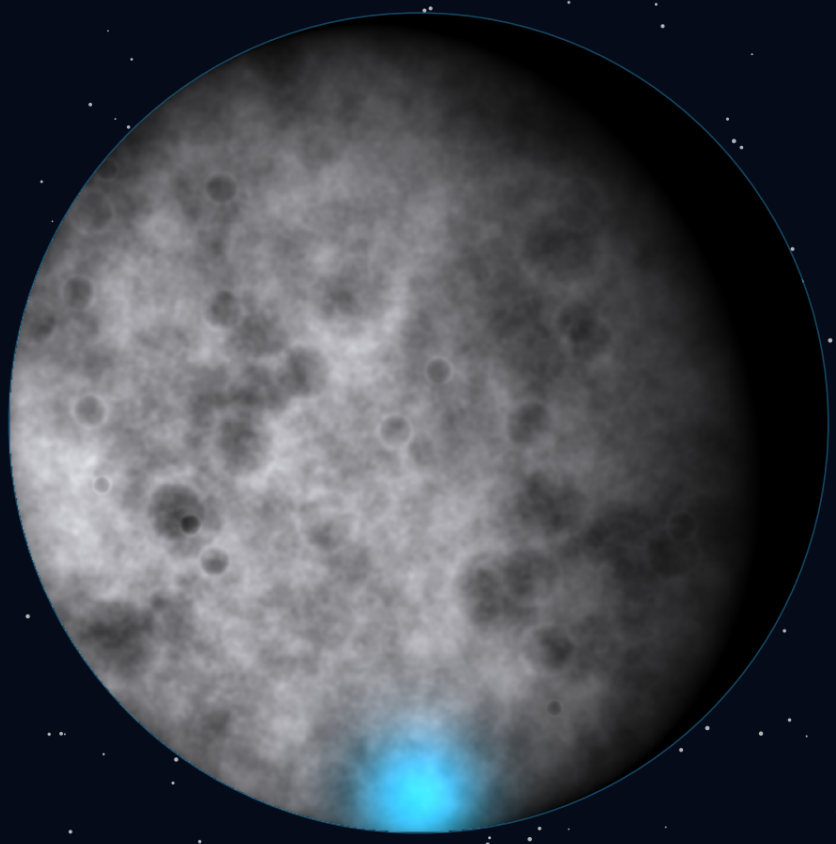


MASTER PLAN

Lunar Ice Detection

A multi-sensor ice-likelihood model for the lunar south pole

DFSAR radar • **OHRC** imagery • **NASA LOLA** topography

Illustrated guide with flowcharts, simulated maps and charts. Maps and charts marked illustrative or simulated are generated examples, not real mission data.

At a glance

This guide turns the master plan into a visual walkthrough. Every section of the original plan is preserved; diagrams and charts are added next to the text to make each idea easier to follow.

#	Section	What it covers
1	Project objective	Goal, three sensors, the CPR/DOP rule
2	Overall architecture	How the data flows into the final map
3	Team division	Five modules and their outputs
4	Phase 1: Data preparation	Study area, coordinate system
5	Phase 2: DFSAR processing	Pipeline, features, radar candidate rule
6	Phase 3: NASA LOLA processing	Layers, physical mask, doubly shadowed craters
7	Phase 4: OHRC processing	Workflow, features, shadow warning
8	Phase 5: Co-registration	Alignment procedure
9	Phase 6: Feature stack	All 28 layers
10	Phase 7: Final model	Physics-informed Random Forest, score classes
11	Labels for training	Positive, negative, uncertain
12	Avoid data leakage	Crater-wise split
13	Candidate extraction	From pixels to ranked candidates
14	Final crater-level score	Weighted score S
15	Validation & sensitivity	Threshold and robustness tests
16	False-positive handling	Reject vs. penalize
17	Final deliverables	Maps, tables, model evidence
18	Presentation statement	What to say, plus final model in one line

The idea in one picture



Final ice likelihood = radar evidence + terrain/shadow suitability + morphology support, minus penalties

1 Project objective

Build a **multi-sensor ice-likelihood model** for the lunar south pole using:

- **DFSAR:** primary evidence for possible subsurface ice.
- **OHRC:** high-resolution surface and morphology validation.
- **NASA LOLA:** terrain, slope, roughness, illumination, and permanent-shadow constraints.

KEY The final output should be called an **ice-probability map**, not a confirmed ice map, because remote sensing provides indirect evidence.

The central scientific rule is:

$$\text{Potential ice candidate} = \text{CPR} > 1 \ \wedge \ \text{DOP} < 0.13$$

ISRO reports that this combination indicates volumetric scattering potentially associated with subsurface ice, especially in doubly shadowed craters [1][2].

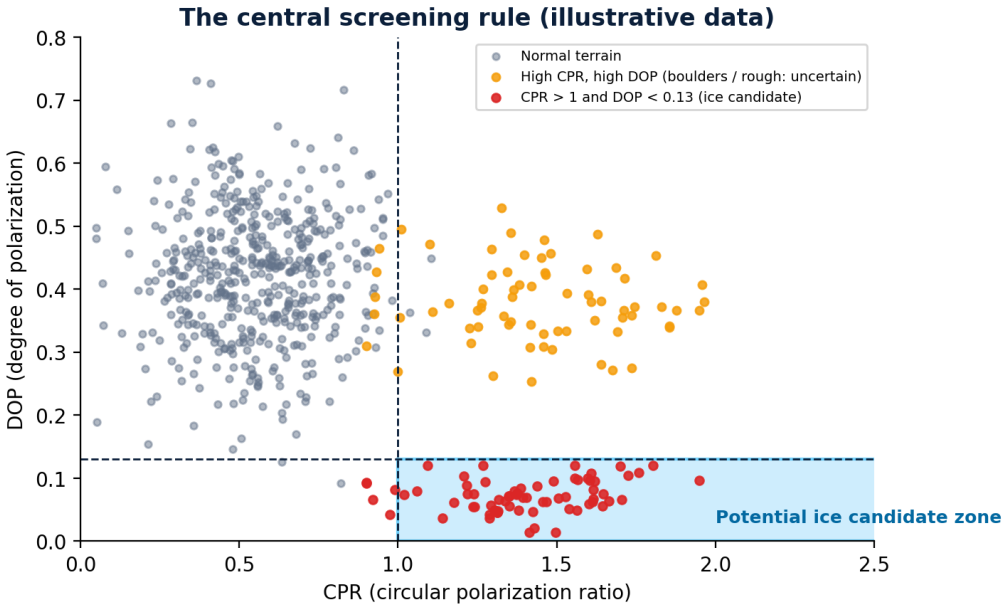


Figure 1. Where the rule sits on a CPR-DOP plot. Boulders can also give high CPR, but their DOP stays high, so they fall outside the ice zone.

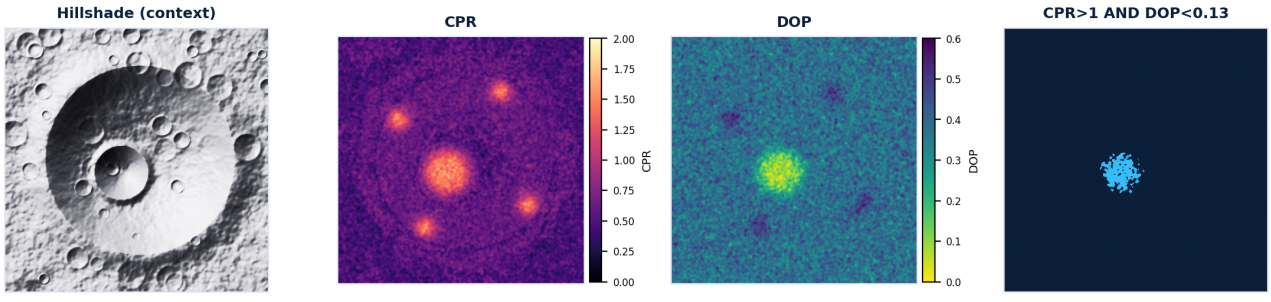


Figure 2. Simulated DFSAR radar view of a doubly shadowed crater: the ice-like signal appears as high CPR plus low DOP inside the inner crater. Bright spots on the rim are boulder fields (high CPR, but high DOP).

2 Overall architecture

The system fuses three sensors. DFSAR finds radar anomalies, LOLA decides whether ice could physically survive there, and OHRC checks what the surface actually looks like. Random Forest fusion then produces the ice-likelihood map and ranked candidates.

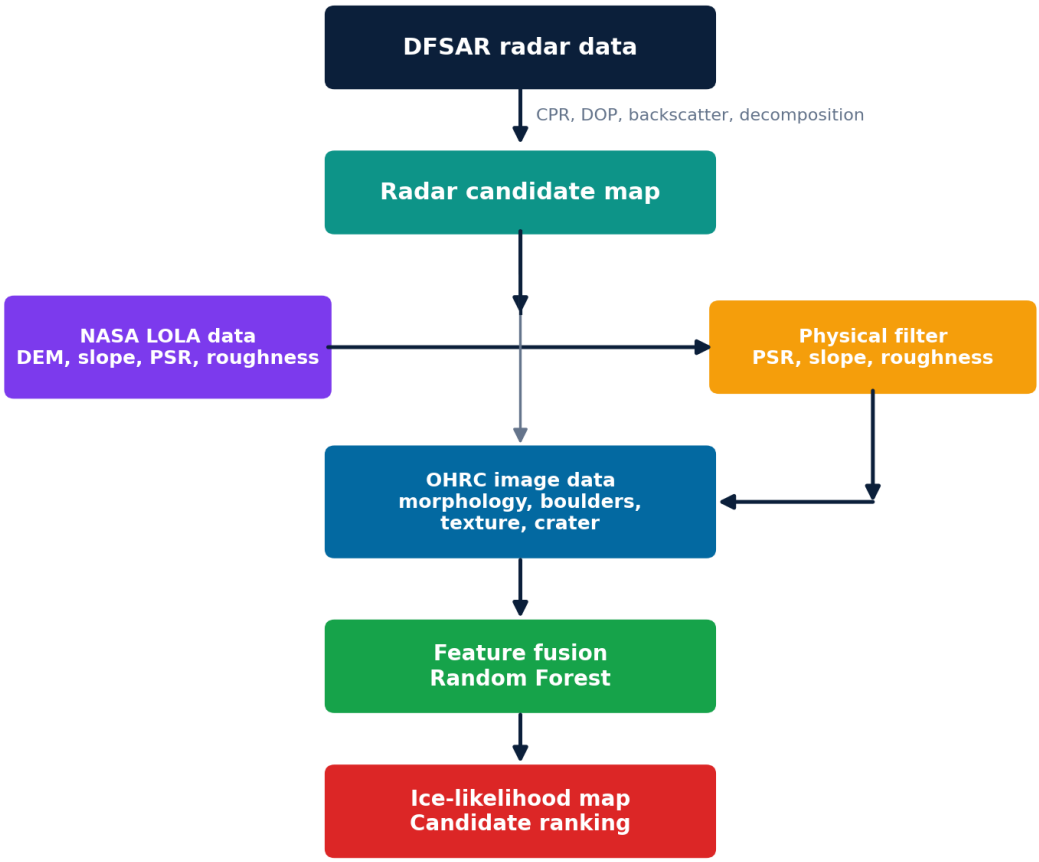


Figure 3. Overall architecture (redrawn from the original diagram).

```
                DFSAR radar data
                       |   CPR, DOP, backscatter, decomposition
                       v
                Radar candidate map
                       |
NASA LOLA data --------+---------> Physical filter
DEM, slope, PSR,       |           PSR, slope, roughness
roughness              v                |
                OHRC image data <-------+
                morphology, boulder,
                texture, crater
                       |
                       v
                Feature fusion / Random Forest
                       |
                       v
                Ice-likelihood map / Candidate ranking
```

3 Team division

Assign the work into five modules.

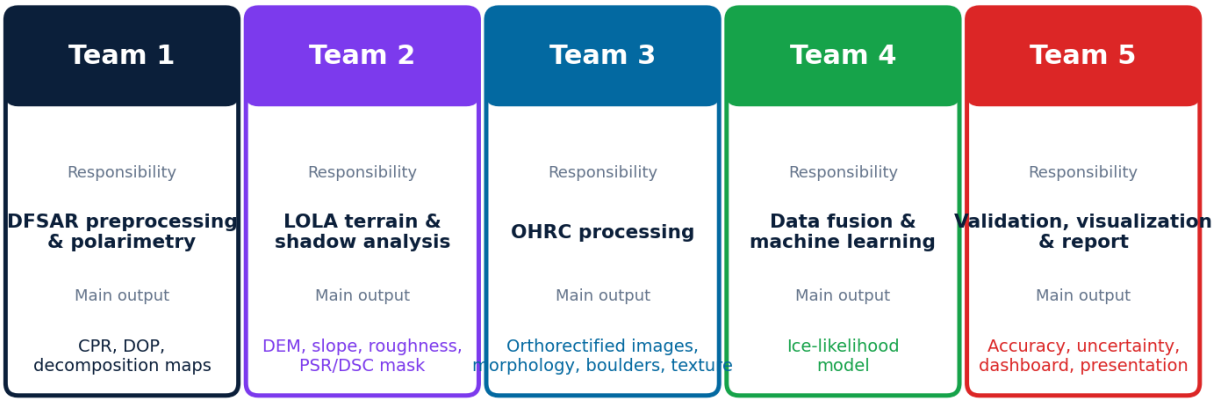


Figure 4. Five teams, one pipeline.

Team	Responsibility	Main output
Team 1	DFSAR preprocessing and polarimetry	CPR, DOP, decomposition maps
Team 2	LOLA terrain and shadow analysis	DEM, slope, roughness, PSR/DSC mask
Team 3	OHRC processing	Orthorectified images, morphology, boulders, texture
Team 4	Data fusion and machine learning	Ice-likelihood model
Team 5	Validation, visualization, and report	Accuracy, uncertainty, dashboard, presentation

RULE All teams must use the **same coordinate system, spatial reference, study area, and metadata format**.

4 Phase 1: Data preparation

4.1 Select the study area

Start with a smaller region where all datasets overlap. Recommended approach:

1. Identify the DFSAR footprint.
2. Find OHRC scenes covering the same area.
3. Select nearby south-polar craters or PSRs.
4. Clip LOLA products to that common area.
5. Use the same geographic boundary for every processing step.

TIP Do not begin with the entire lunar south pole. First build and test the workflow on **one or two craters**, then scale it.

4.2 Coordinate system

Use a lunar south-polar stereographic projection for all data. Every dataset must be:

- Reprojected.
- Co-registered.

- Resampled to a common grid.
- Stored with the same pixel origin and extent.

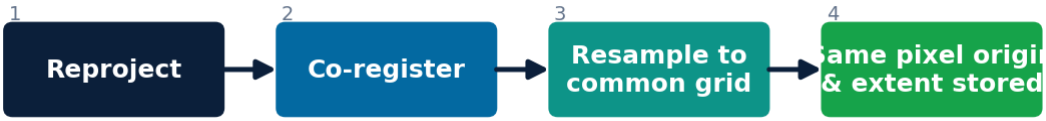


Figure 5. Bringing every dataset onto one grid.

The final analytical grid should be based on the spatial resolution of **DFSAR, not OHRC**. OHRC can be displayed at high resolution, but it should not create false precision beyond the radar resolution.

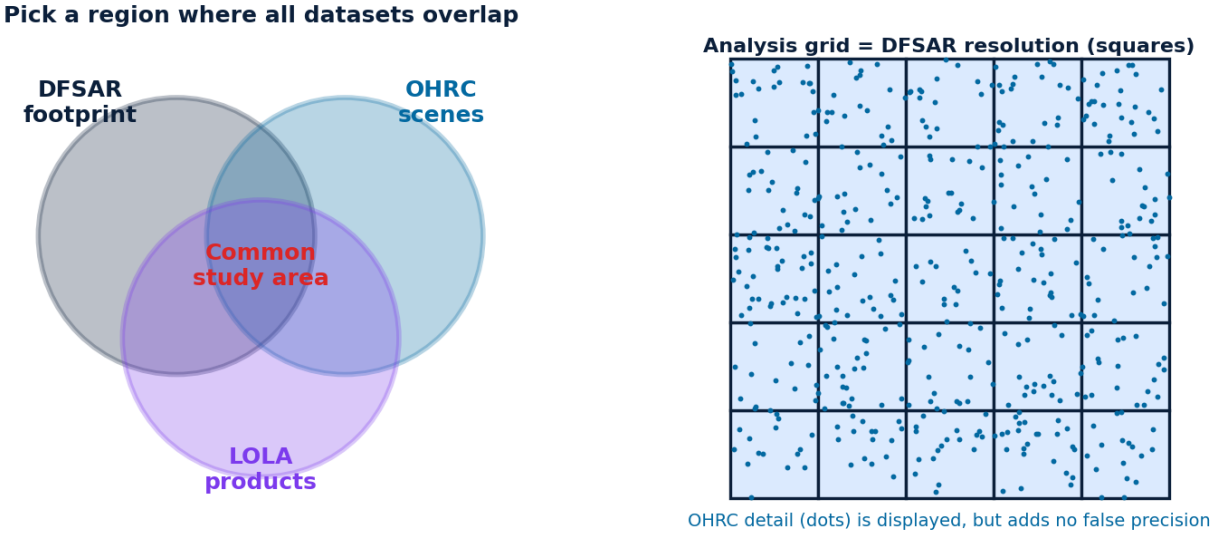


Figure 6. Left: the study area is where all three datasets overlap. Right: the analysis grid follows DFSAR resolution; OHRC detail does not add precision.

5 Phase 2: DFSAR processing

DFSAR is the main detection dataset because it provides radar observations and polarimetric information about the surface and shallow subsurface.

5.1 Processing sequence

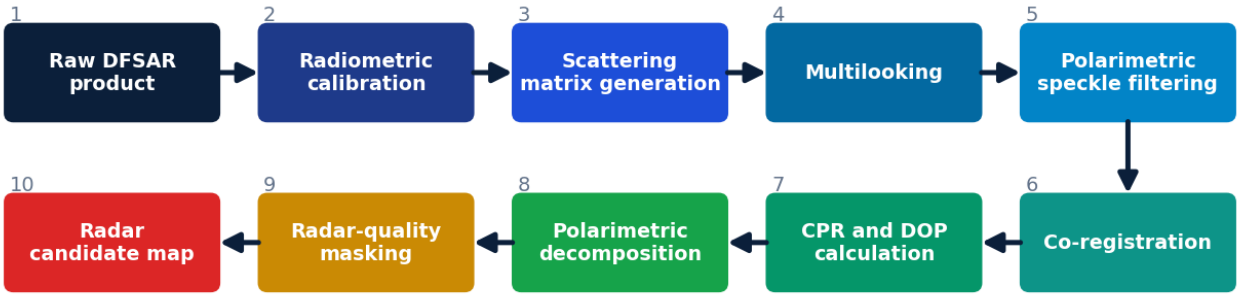


Figure 7. DFSAR processing chain (read left to right, then right to left on the second row).

```
Raw DFSAR product -> Radiometric calibration -> Scattering matrix generation
-> Multilooking -> Polarimetric speckle filtering -> Co-registration
-> CPR and DOP calculation -> Polarimetric decomposition
-> Radar-quality masking -> Radar candidate map
```

5.2 Features to calculate

Generate these layers:

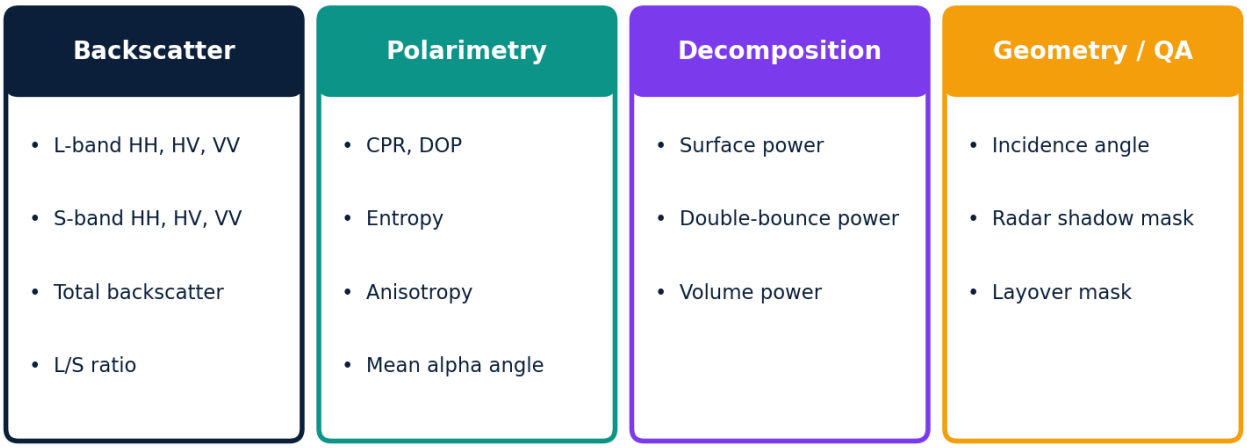


Figure 8. The 19 DFSAR layers, grouped.

- L-band HH.
- L-band HV.
- L-band VV.
- S-band HH.
- S-band HV.
- S-band VV.
- Total backscatter.
- CPR.
- DOP.
- L-band/S-band ratio.

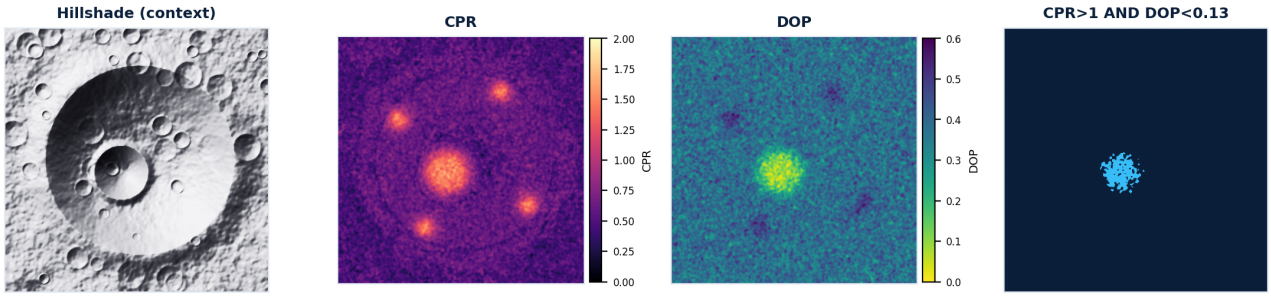
- Entropy.
- Anisotropy.
- Mean alpha angle.
- Surface-scattering power.
- Double-bounce-scattering power.
- Volume-scattering power.
- Incidence angle.
- Radar shadow mask.
- Layover mask.

5.3 Radar candidate rule

Create the first candidate mask:

```
radar_candidate = (
    (CPR > 1.0) &
    (DOP < 0.13) &
    (volume_scattering > selected_threshold) &
    (radar_quality == 1)
)
```

The CPR/DOP criterion should be treated as a **strong screening rule, not as final confirmation**. ISRO's analysis used this combination to identify possible subsurface ice signatures in four doubly shadowed south-polar craters [1].



Simulated DFSAR radar products over a doubly shadowed crater (illustrative)

Figure 9. The candidate mask is the intersection of the CPR and DOP conditions (simulated).

6 Phase 3: NASA LOLA processing

NASA LOLA data is open through NASA's Planetary Data System and related planetary GIS archives [3].

6.1 Required LOLA layers

Download:



Figure 10. The six LOLA products to download.

- **LDEM:** elevation.
- **LDSM:** slope.
- **LDRM:** roughness.
- **LPSR:** permanent-shadow mask.
- **LDEC:** laser-observation count.
- **DEM error:** elevation uncertainty.

NASA's south-polar LOLA products include DEMs, slope maps, roughness maps, PSR maps, and data-quality layers [3][4].

6.2 LOLA processing

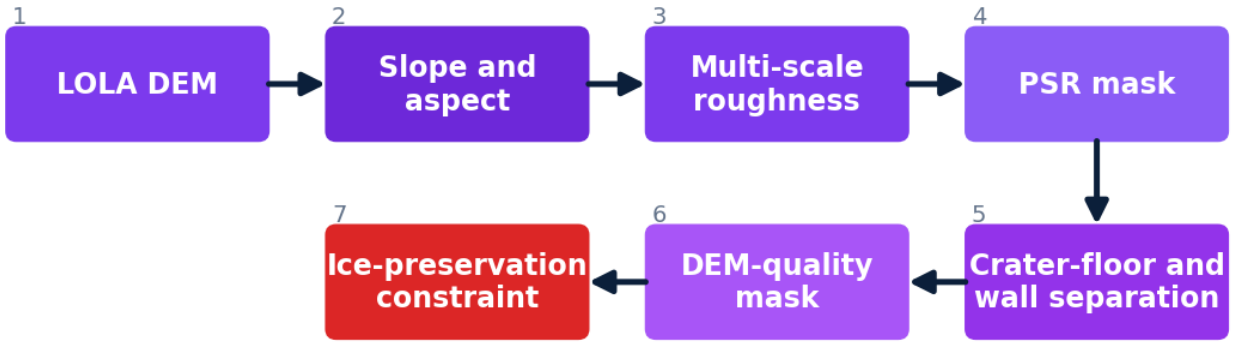


Figure 11. LOLA processing chain.

```
LOLA DEM -> Slope and aspect -> Multi-scale roughness -> PSR mask
-> Crater-floor and wall separation -> DEM-quality mask
-> Ice-preservation constraint
```

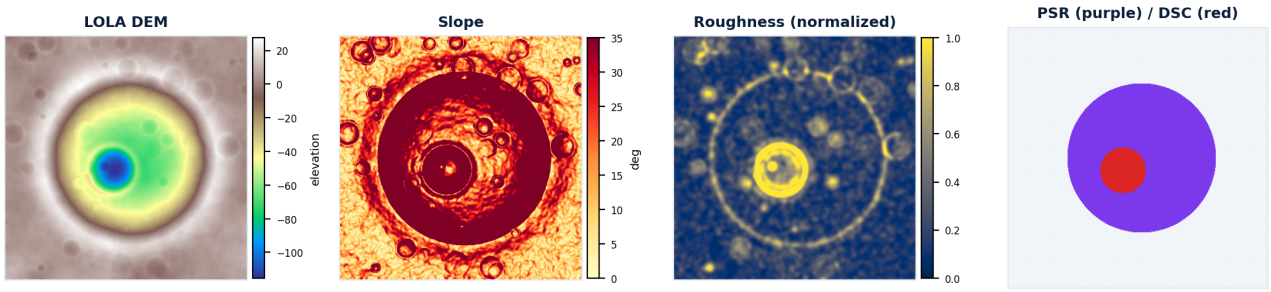


Figure 12. Simulated LOLA-derived layers for the same crater.

6.3 Create the physical plausibility mask

```
physical_mask = (
    (PSR == 1) &
    (DEM_error < selected_limit) &
    (laser_count >= selected_minimum)
)
```

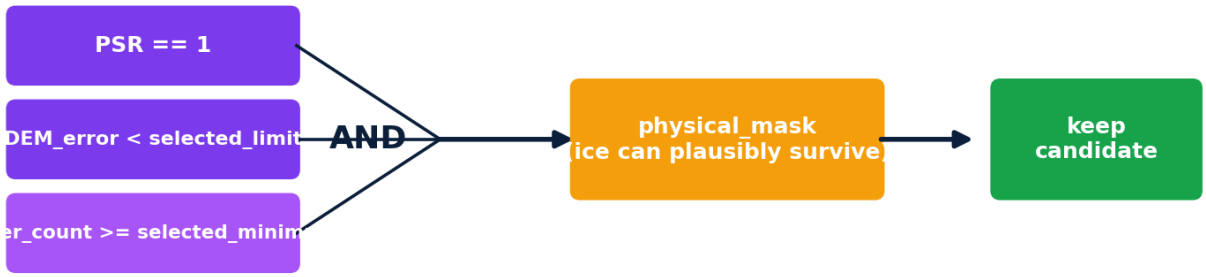


Figure 13. The physical plausibility mask: all three conditions must hold.

Do not automatically discard all high-roughness regions. Instead, use roughness as a **confidence penalty** because subsurface ice may exist beneath rough regolith.

Use:

```
roughness_penalty = normalize(roughness)
```

rather than:

```
roughness == 0
```

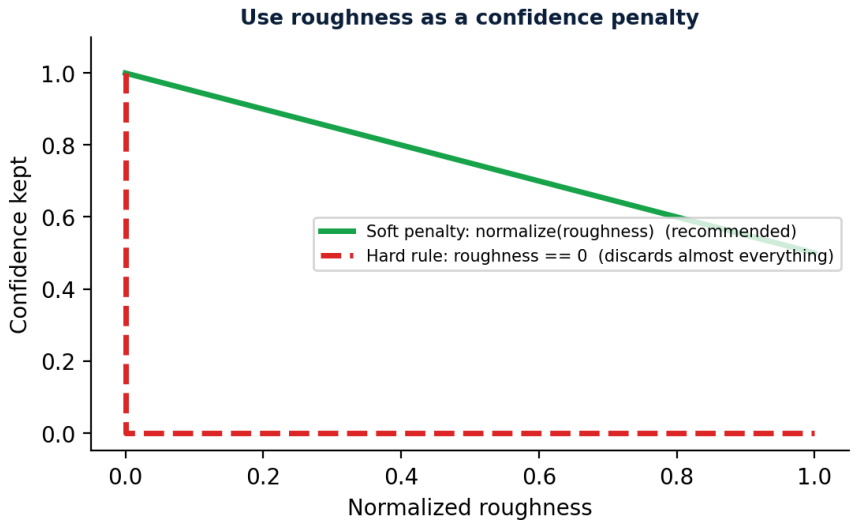


Figure 14. A soft penalty keeps rough-but-plausible areas alive; a hard rule would discard almost everything.

6.4 Detect doubly shadowed craters

A doubly shadowed crater is a strong target because it is inside a permanently shadowed region and has additional local shielding.

Use:

```
DSC = (
    (PSR == 1) &
    (crater_floor == 1) &
    (local_shadowing == 1)
)
```

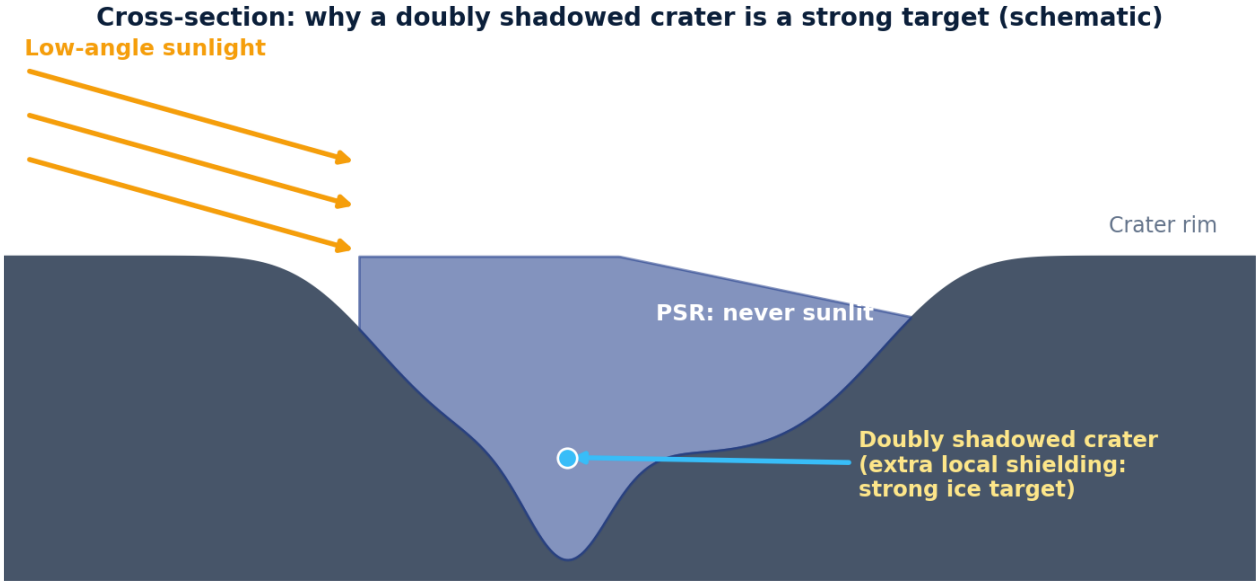


Figure 15. A small crater inside a permanently shadowed crater is shielded twice.

If a published DSC layer is unavailable, estimate it from:

- LOLA DEM.
- Crater geometry.
- Local horizon obstruction.
- Illumination simulation.
- OHRC shadow boundaries.

7 Phase 4: OHRC processing

OHRC should **support and validate** the DFSAR result. It should not be used as the only ice detector.

OHRC has approximately 0.25–0.32 m/pixel imaging capability, making it useful for detailed crater morphology and boulder analysis [5][6].

7.1 OHRC workflow

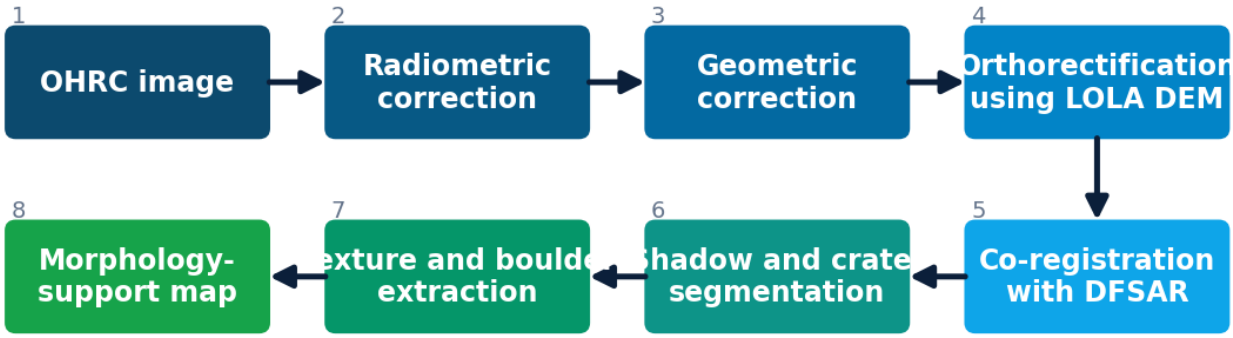


Figure 16. OHRC processing chain.

```
OHRC image -> Radiometric correction -> Geometric correction
-> Orthorectification using LOLA DEM -> Co-registration with DFSAR
-> Shadow and crater segmentation -> Texture and boulder extraction
-> Morphology-support map
```

7.2 Extract these OHRC features

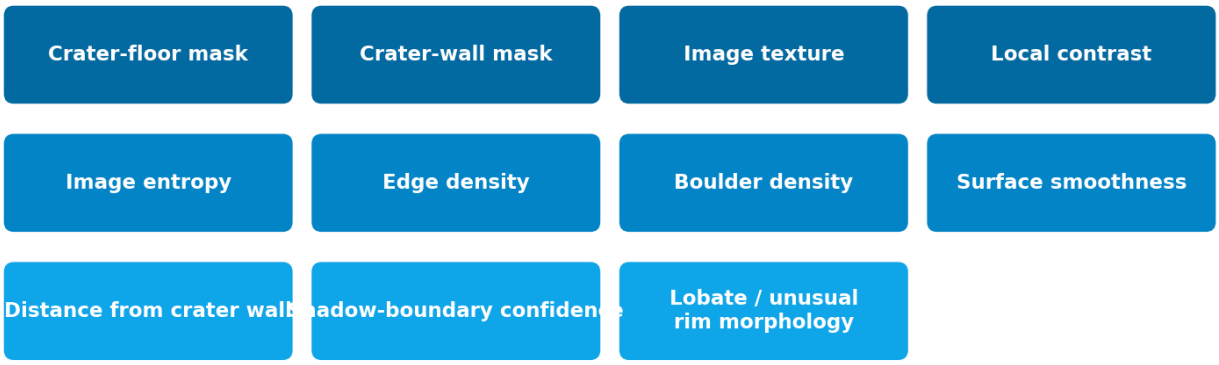


Figure 17. The eleven OHRC features.

- Crater-floor mask.
- Crater-wall mask.
- Image texture.
- Local contrast.
- Image entropy.
- Edge density.
- Boulder density.
- Surface smoothness.
- Distance from crater wall.
- Shadow-boundary confidence.

- Possible lobate or unusual rim morphology.

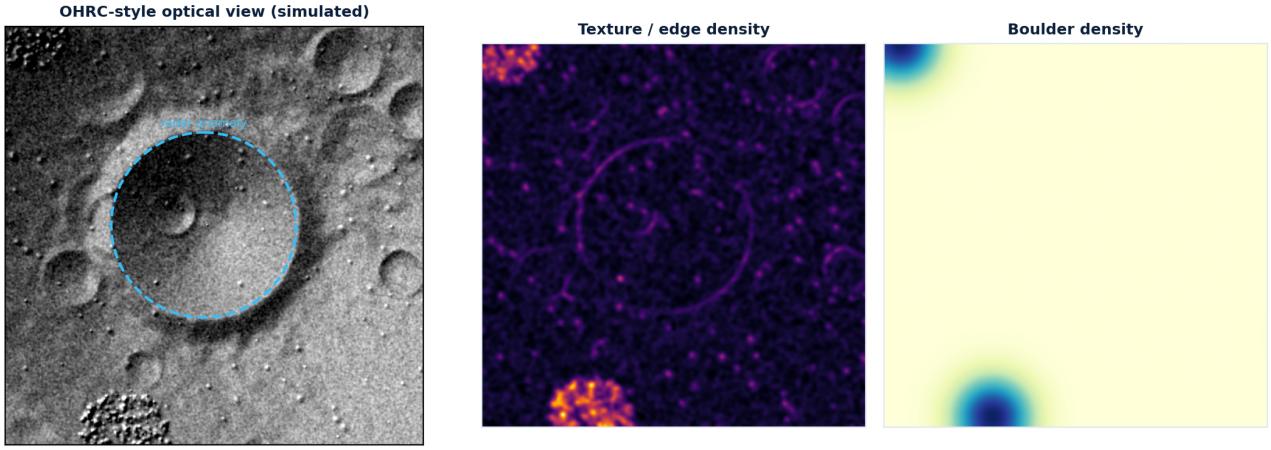


Figure 18. OHRC-style view (simulated at fine resolution): boulders and texture help decide whether a radar anomaly sits on a smooth floor or a rocky area. The dashed circle marks the radar anomaly.

WARN **Important:** a dark OHRC region is not automatically ice. It may simply be in shadow. Use OHRC mainly to determine whether a radar anomaly occurs on the surface types below.

- A smooth crater floor.
- A rocky boulder field.
- A steep crater wall.
- A shadow boundary.
- A region contaminated by geometric effects.

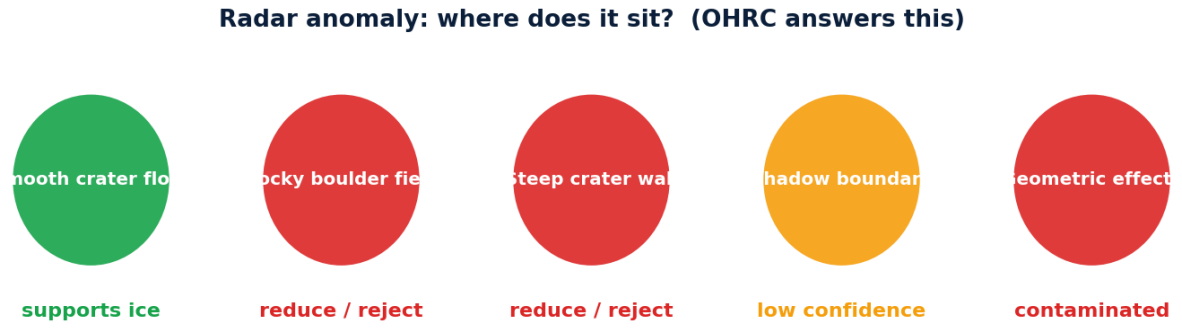


Figure 19. Same radar signal, different meaning depending on the surface.

8 Phase 5: Co-registration

This is one of the most important technical stages. Misregistration between DFSAR, OHRC, and LOLA can create false overlap and incorrect model results.

Co-registration procedure

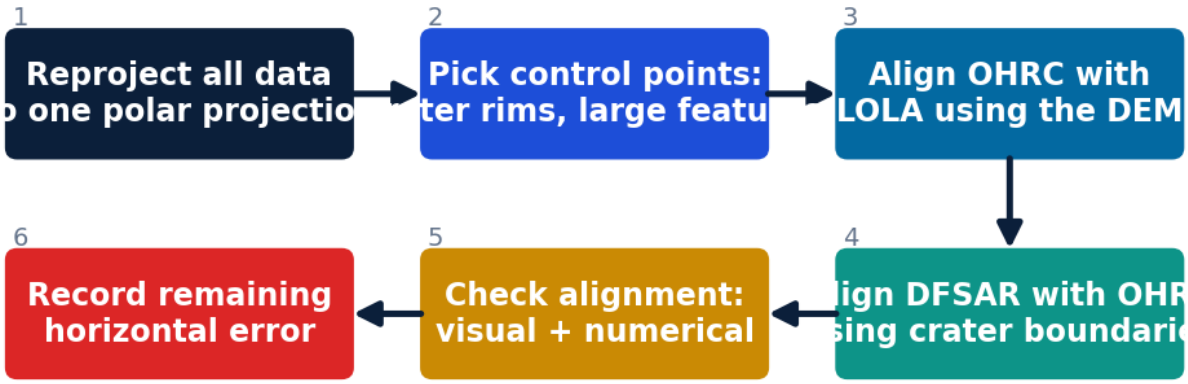


Figure 20. Six steps to a trustworthy alignment.

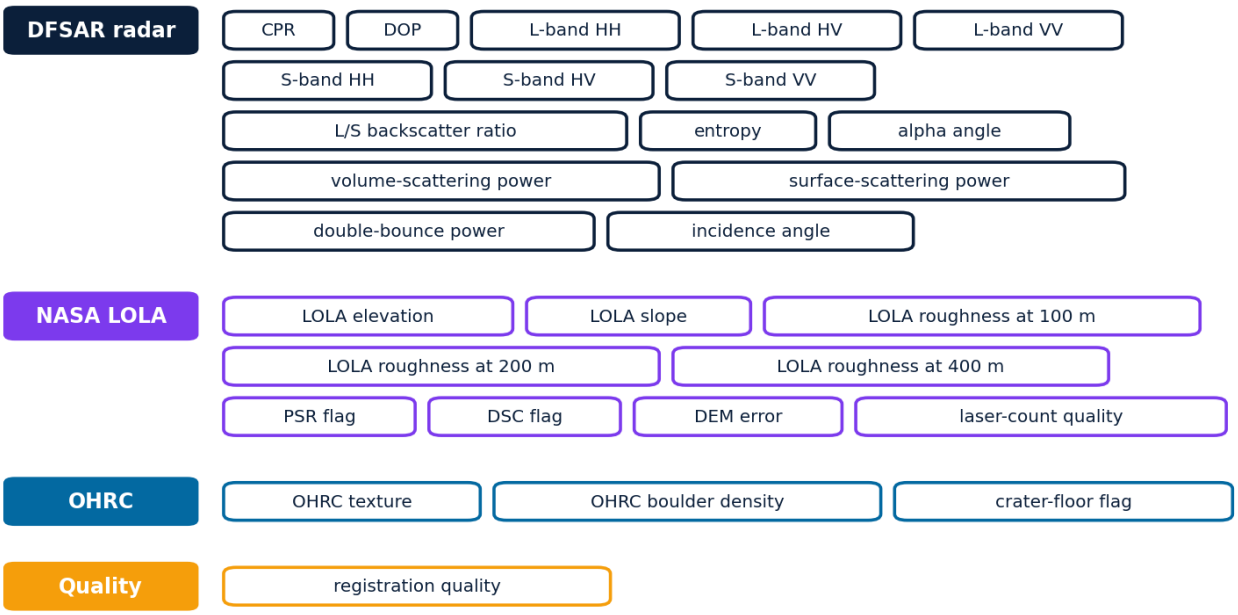
1. Reproject all data to the same polar projection.
2. Use crater rims and large topographic features as control points.
3. Align OHRC with LOLA using the DEM.
4. Align DFSAR with OHRC using crater boundaries.
5. Check alignment visually and numerically.
6. Record the remaining horizontal error.

Use only regions with acceptable registration error. Store this as a feature:

```
registration_quality
```

9 Phase 6: Create the feature stack

Create one multiband raster or table in which each pixel or cell contains:



28 layers per pixel / cell -> input to the final model

Figure 21. The 28-layer feature stack, color-coded by source.

```
CPR
DOP
L-band HH
L-band HV
L-band VV
S-band HH
S-band HV
S-band VV
L/S backscatter ratio
entropy
alpha angle
volume-scattering power
surface-scattering power
double-bounce power
LOLA elevation
LOLA slope
LOLA roughness at 100 m
LOLA roughness at 200 m
LOLA roughness at 400 m
PSR flag
DSC flag
OHRC texture
OHRC boulder density
crater-floor flag
DEM error
laser-count quality
incidence angle
registration quality
```

This is the input to the final model.

10 Phase 7: Final model

Recommended model

Use a **physics-informed Random Forest** first.

Reasons:

- It works with mixed radar, terrain, and image features.
- It does not require a very large labelled image dataset.
- It is easier to explain than a deep neural network.
- Feature importance can be reported.
- It can produce a probability-like confidence score.

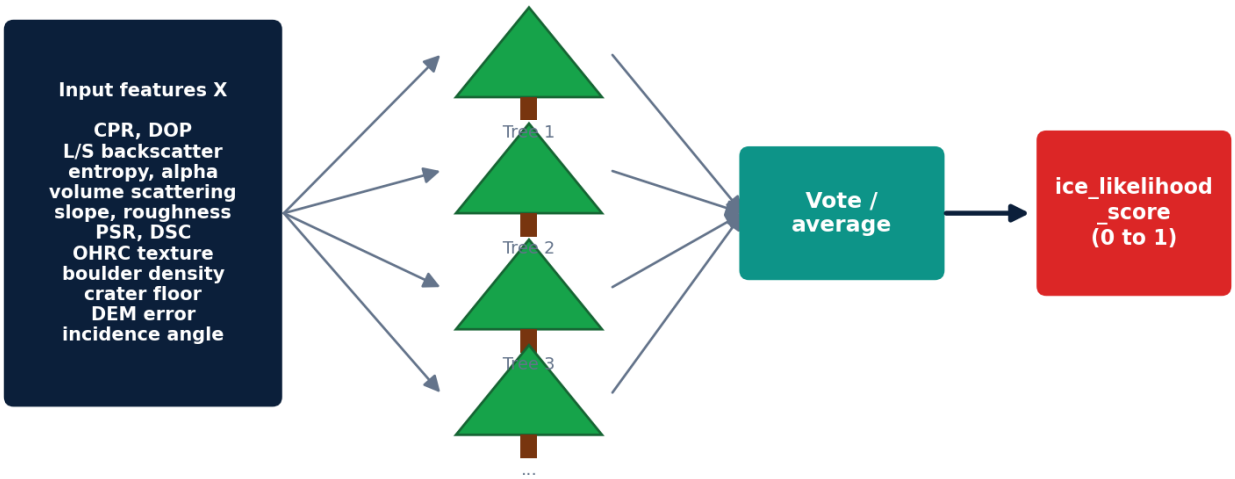


Figure 22. Many decision trees vote on the same pixel; the average becomes the ice-likelihood score.

Model input

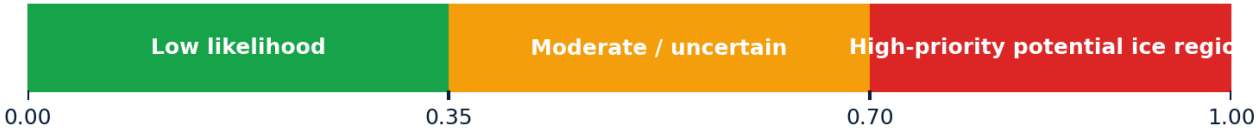
```
X = [
    CPR, DOP,
    L_band_backscatter,
    S_band_backscatter,
    entropy,
    alpha,
    volume_scattering,
    slope,
    roughness,
    PSR,
    DSC,
    OHRC_texture,
    boulder_density,
    crater_floor,
    DEM_error,
    incidence_angle
]
```

Model output

The model should produce:

```
ice_likelihood_score
```

Use three final categories:



Project confidence classes, NOT measured ice concentration

Figure 23. Score classes.

Score	Interpretation
0.00–0.35	Low likelihood
0.35–0.70	Moderate or uncertain likelihood
0.70–1.00	High-priority potential ice region

These values are **project confidence classes, not measured ice concentration**.

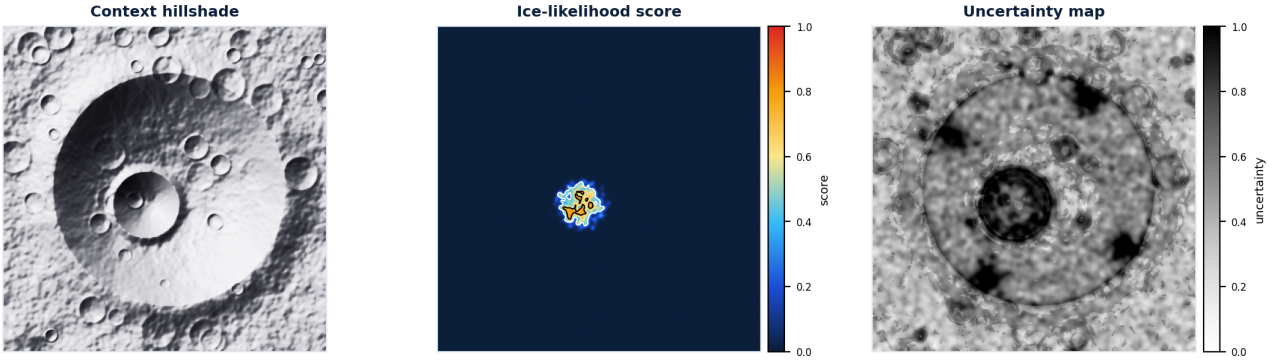


Figure 24. Example output (simulated): the high-likelihood core sits in the shadowed inner crater; the uncertainty map flags boulder fields and rough walls.

11 Labels for training

Because confirmed pixel-level ice labels are unavailable, use **weak labels**.

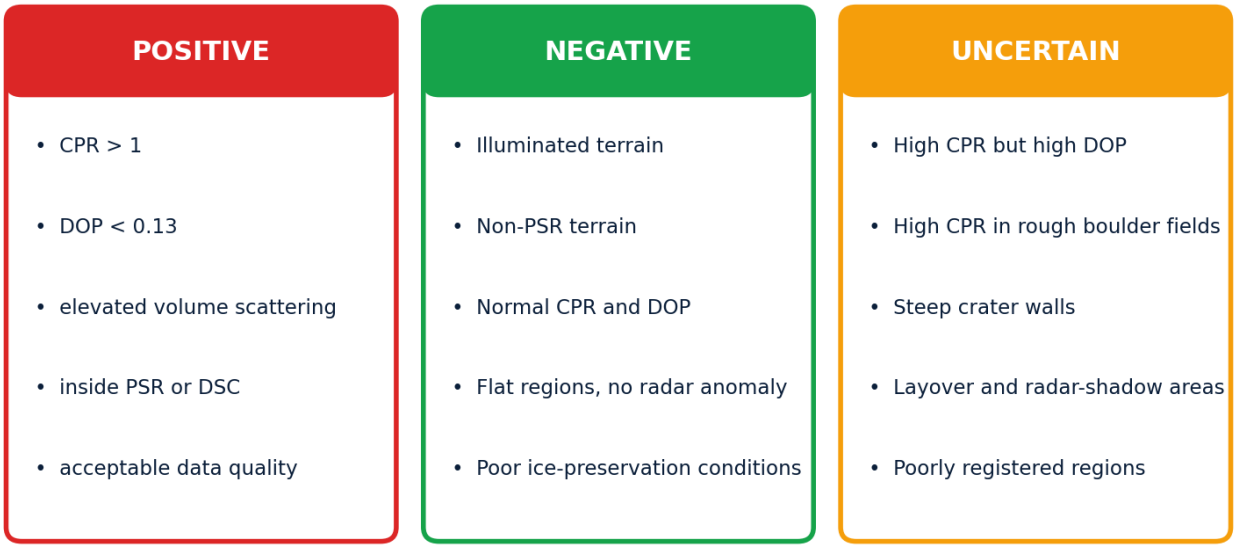


Figure 25. Three label classes.

Positive samples

Select samples that satisfy:

```
CPR > 1
DOP < 0.13
elevated volume scattering
inside PSR or DSC
acceptable data quality
```

Negative samples

Select:

- Illuminated terrain.
- Non-PSR terrain.
- Areas with normal CPR and DOP.
- Flat regions without radar anomalies.
- Areas with poor ice-preservation conditions.

Uncertain samples

Keep a third class for:

- High CPR but high DOP.
- High CPR in rough boulder fields.
- Steep crater walls.
- Layover and radar-shadow areas.
- Poorly registered regions.

RULE	Do not force uncertain samples into the positive or negative class.

12 Avoid data leakage

Do not randomly split neighboring pixels into training and testing data. Neighboring pixels are highly similar, which will produce misleadingly high accuracy.

Instead:

```
Training: selected craters
Validation: different craters
Testing: completely separate crater
```

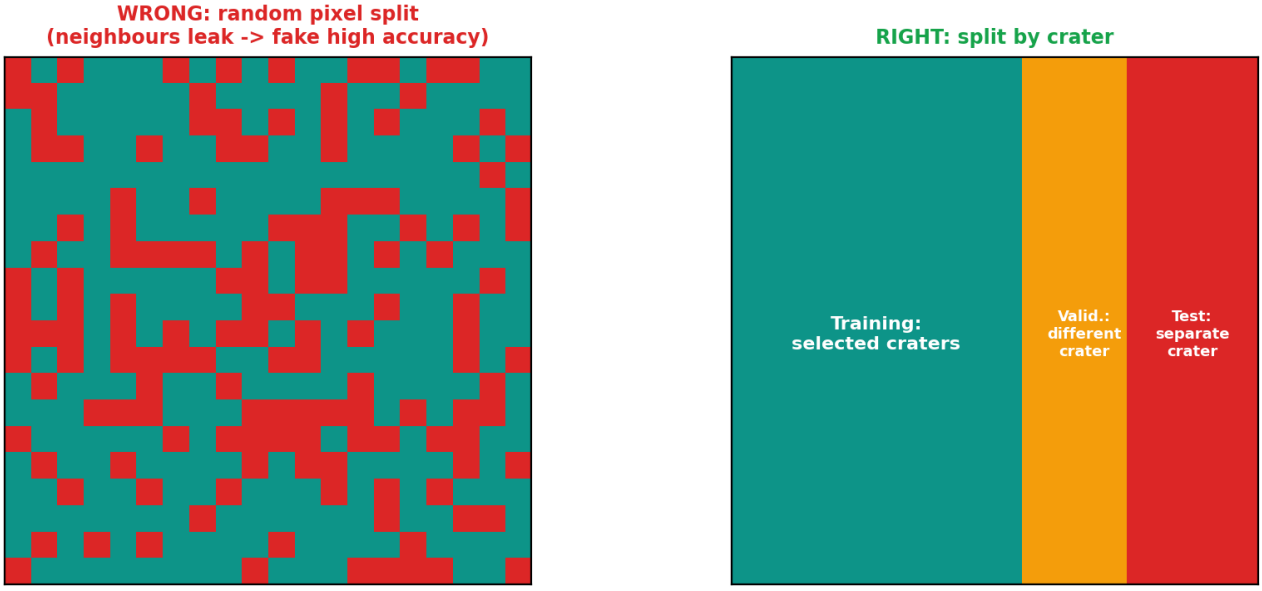


Figure 26. Split by crater, never by pixel.

Example:

```
Train on Faustini and Shackleton
Validate on Haworth
Test on another south-polar crater
```

The exact craters should depend on your data coverage.

13 Candidate extraction

After generating the pixel-level probability map:

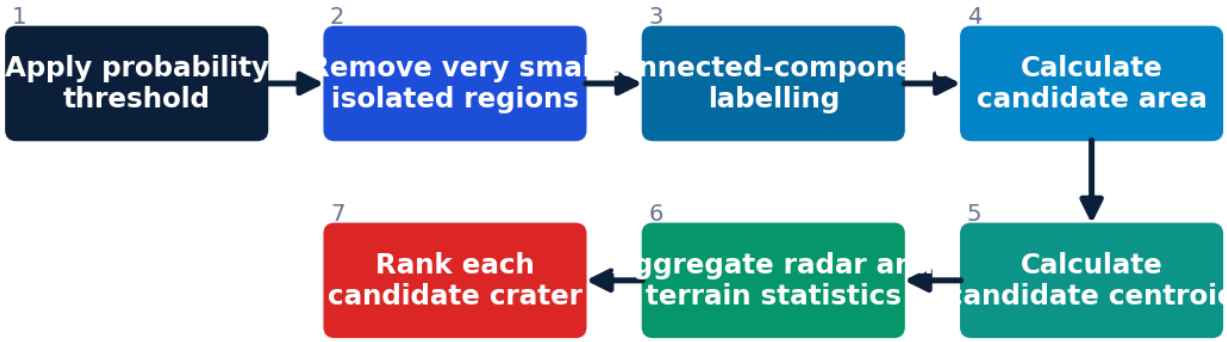


Figure 27. From probability map to ranked list.

1. Apply a probability threshold.
2. Remove very small isolated regions.
3. Apply connected-component labelling.
4. Calculate candidate area.
5. Calculate candidate centroid.
6. Aggregate radar and terrain statistics.
7. Rank each candidate crater.

Candidate ranking table

Create columns for:

Column	Column	Column
Candidate ID	Crater name	Latitude
Longitude	Area	Median CPR
Median DOP	Volume-scattering percentage	PSR percentage
DSC flag	Median slope	Median roughness
OHRC boulder density	DEM error	Model probability
Uncertainty	Final rank	

14 Final crater-level score

Use the pixel probability together with physical constraints:

$$S = 0.30R + 0.20P + 0.15V + 0.15T + 0.10O + 0.10Q$$

where:

- R: radar evidence.
- P: PSR/DSC evidence.
- V: volume-scattering evidence.
- T: terrain suitability.
- O: OHRC morphology support.
- Q: data quality.

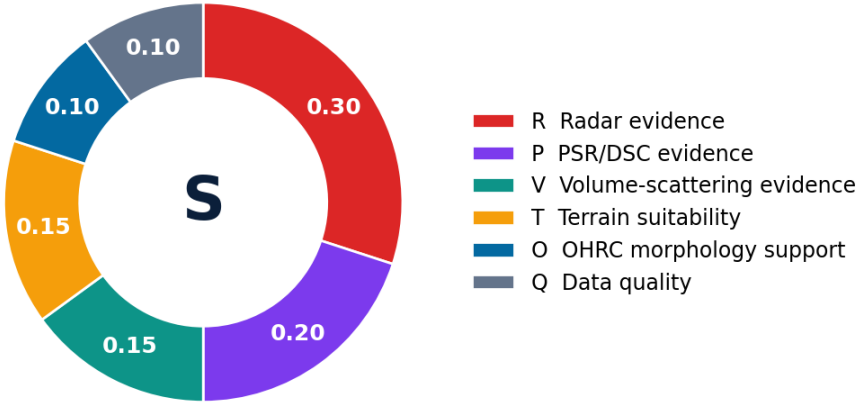


Figure 28. Weight of each evidence term in the score S.

RULE	A candidate should be ranked high only when it has **both** radar evidence and a physically suitable location.

15 Validation and sensitivity testing

Run the complete model several times with different thresholds:

```
CPR > 0.8, 1.0, 1.2
DOP < 0.13, 0.20, 0.30
```

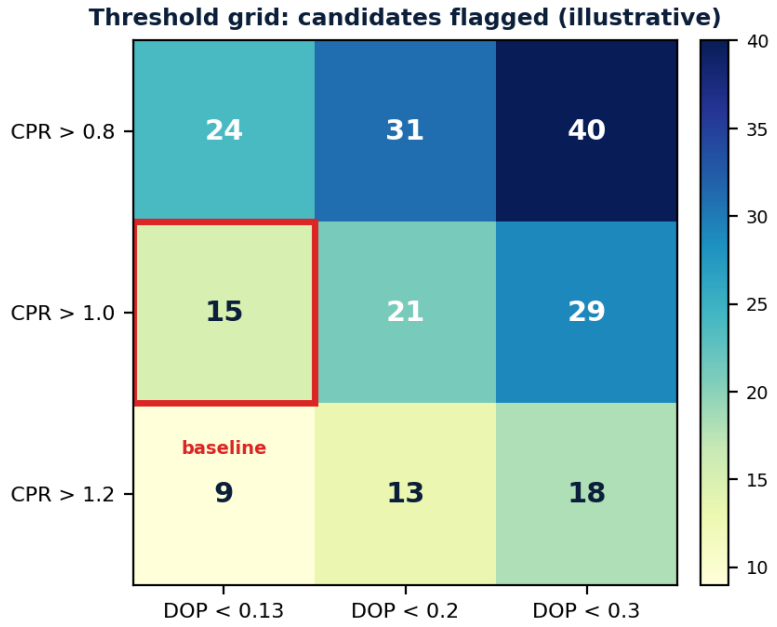


Figure 29. Nine threshold combinations (numbers are illustrative). Stable candidates should survive most of them.

Also test:

- Different speckle-filter window sizes.
- Different roughness penalties.
- Different probability thresholds.
- Small image-registration shifts.
- L-band only versus L-band + S-band.
- DFSAR only versus DFSAR + LOLA.
- DFSAR + LOLA versus all three datasets.

GOAL The final candidates should remain reasonably stable under these tests.

16 False-positive handling

Your model must explicitly reject or reduce confidence for:

Risk	Why it fools the model	Action
High CPR caused by boulders	Rocks scatter radar strongly, mimicking ice	Reduce confidence
Rough crater walls	Rough slopes raise CPR	Reduce confidence
Double-bounce scattering	Corner-like reflections raise CPR	Down-weight
Radar layover	Geometry distortion on steep slopes	Reject
Radar shadow	No valid radar return	Reject
Steep slopes	Unreliable geometry, no stable ice	Penalize

Risk	Why it fools the model	Action
Poor DEM coverage	Terrain and shadow data untrustworthy	Reject (DEM error)
Poor DFSAR–OHRC alignment	Layers do not truly overlap	Reject
OHRC optical darkness caused only by shadow	Dark does not mean ice	Ignore as evidence
Noise and speckle	Random bright pixels	Filter
Isolated high-CPR pixels	Single pixels are not a patch	Remove small regions

Recommended rejection rule:

```
reject_candidate = (
    radar_shadow == 1
    or layover == 1
    or registration_quality == 0
    or DEM_error > limit
)
```



Figure 30. Hard rejects versus soft penalties.

For roughness and boulders, **reduce probability** rather than always rejecting.

17 Final deliverables

Your team should submit the following:

Maps

1. DFSAR CPR map.
2. DFSAR DOP map.
3. DFSAR volume-scattering map.
4. LOLA DEM.
5. LOLA slope map.
6. LOLA roughness map.
7. PSR/DSC map.
8. OHRC morphology map.
9. Final ice-likelihood map.
10. Uncertainty map.

Tables

1. Ranked crater candidates.
2. Feature statistics for each candidate.
3. Model performance.
4. Threshold-sensitivity results.
5. False-positive examples.

Model evidence

- Confusion matrix, if labels permit.
- ROC-AUC or precision–recall, with limitations clearly stated.
- Feature importance.
- Crater-level validation.
- Uncertainty analysis.

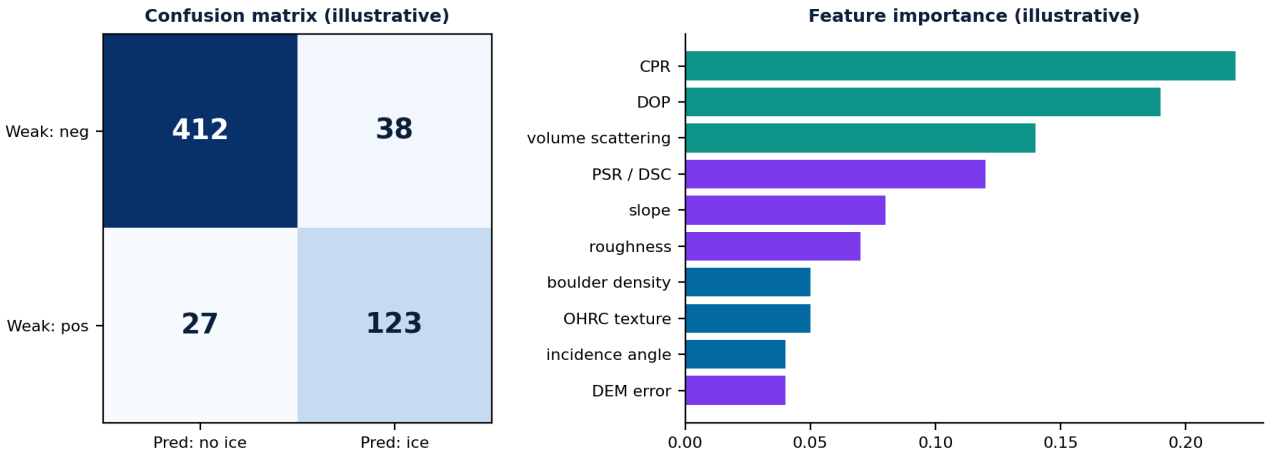


Figure 31. Example evidence panel (illustrative numbers). Weak labels mean these metrics need clear caveats.

18 Recommended final presentation statement

Use this statement in your presentation:

> *“Our system does not depend on a single image or a single threshold. It first uses DFSAR polarimetric evidence to identify radar anomalies, then uses NASA LOLA topography and permanent-shadow information to determine whether ice preservation is physically plausible, and finally uses OHRC imagery to reject rough, rocky, or geometrically ambiguous regions. The output is a ranked ice-likelihood map for lunar south-polar exploration.”*

Final model in one line

```
Final ice likelihood =
DFSAR radar evidence
+ LOLA PSR and terrain suitability
+ OHRC morphology support
− roughness, boulder, geometry, and data-quality penalties
```



Figure 32. The final model in one picture.

The best implementation order is:

```
Start with:
DFSAR CPR + DOP + LOLA PSR

Then add:
DFSAR decomposition + LOLA roughness + slope

Finally add:
OHRC morphology + Random Forest + uncertainty mapping
```

Working baseline early, advanced features added without redesigning the system



Figure 33. Staged implementation roadmap.

This staged approach gives your team a working baseline early, while allowing more advanced features to be added without redesigning the entire system.

References [1]–[6] are cited as in the original plan.